

Finite Precision Gram-Schmidt Walks

ACO Student Seminar Talk

Peter Chen, Joint work with Emile Anand and Jan van den Brand

2026-10-02

Discrepancy Theory



Discrepancy on Set Families

- Informally, split some structure in a balanced way

Discrepancy on Set Families

- Informally, split some structure in a balanced way
- Fix some $\mathcal{F} \subseteq 2^{[n]}$ and let $\chi : [n] \rightarrow \{-1, 1\}$ be a two coloring

Discrepancy on Set Families

- Informally, split some structure in a balanced way
- Fix some $\mathcal{F} \subseteq 2^{[n]}$ and let $\chi : [n] \rightarrow \{-1, 1\}$ be a two coloring
- Let R, B denote the set of units for each color, so $R = \chi^{-1}(-1), B = \chi^{-1}(1)$

Discrepancy on Set Families

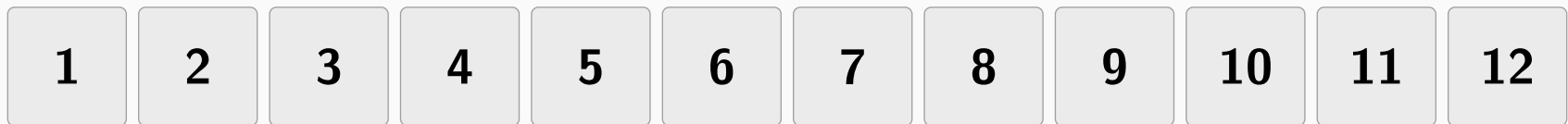
- Informally, split some structure in a balanced way
- Fix some $\mathcal{F} \subseteq 2^{[n]}$ and let $\chi : [n] \rightarrow \{-1, 1\}$ be a two coloring
- Let R, B denote the set of units for each color, so $R = \chi^{-1}(-1)$, $B = \chi^{-1}(1)$
- We say the discrepancy of $\mathcal{F}$ with respect to a coloring χ is

$$\text{disc}(\mathcal{F}, \chi) = \max_{S \in \mathcal{F}} ||S \cap R| - |S \cap B||$$

Discrepancy on Set Families

- Informally, split some structure in a balanced way
- Fix some $\mathcal{F} \subseteq 2^{[n]}$ and let $\chi : [n] \rightarrow \{-1, 1\}$ be a two coloring
- Let R, B denote the set of units for each color, so $R = \chi^{-1}(-1), B = \chi^{-1}(1)$
- We say the discrepancy of $\mathcal{F}$ with respect to a coloring χ is

$$\text{disc}(\mathcal{F}, \chi) = \max_{S \in \mathcal{F}} ||S \cap R| - |S \cap B||$$



The universe $[12]$

Discrepancy on Set Families

- Informally, split some structure in a balanced way
- Fix some $\mathcal{F} \subseteq 2^{[n]}$ and let $\chi : [n] \rightarrow \{-1, 1\}$ be a two coloring
- Let R, B denote the set of units for each color, so $R = \chi^{-1}(-1), B = \chi^{-1}(1)$
- We say the discrepancy of $\mathcal{F}$ with respect to a coloring χ is

$$\text{disc}(\mathcal{F}, \chi) = \max_{S \in \mathcal{F}} ||S \cap R| - |S \cap B||$$

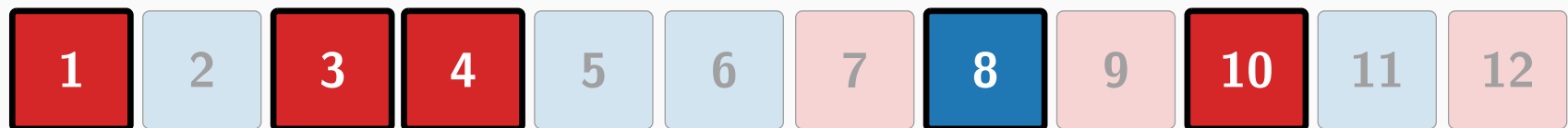


A coloring $\chi : [12] \rightarrow \{\text{R}, \text{B}\}$
 $\mathcal{F} = \{\{1, 3, 4, 8, 10\}, \{5, 6, 7, 8\}, [12]\}$

Discrepancy on Set Families

- Informally, split some structure in a balanced way
- Fix some $\mathcal{F} \subseteq 2^{[n]}$ and let $\chi : [n] \rightarrow \{-1, 1\}$ be a two coloring
- Let R, B denote the set of units for each color, so $R = \chi^{-1}(-1), B = \chi^{-1}(1)$
- We say the discrepancy of $\mathcal{F}$ with respect to a coloring χ is

$$\text{disc}(\mathcal{F}, \chi) = \max_{S \in \mathcal{F}} ||S \cap R| - |S \cap B||$$



$$S_1 = \{1, 3, 4, 8, 10\} : \quad |S_1 \cap R| = 4, \quad |S_1 \cap B| = 1, \quad |4 - 1| = 3$$

Discrepancy on Set Families

- Informally, split some structure in a balanced way
- Fix some $\mathcal{F} \subseteq 2^{[n]}$ and let $\chi : [n] \rightarrow \{-1, 1\}$ be a two coloring
- Let R, B denote the set of units for each color, so $R = \chi^{-1}(-1), B = \chi^{-1}(1)$
- We say the discrepancy of $\mathcal{F}$ with respect to a coloring χ is

$$\text{disc}(\mathcal{F}, \chi) = \max_{S \in \mathcal{F}} ||S \cap R| - |S \cap B||$$



$$S_2 = \{5, 6, 7, 8\} : \quad |S_2 \cap R| = 1, \quad |S_2 \cap B| = 3, \quad |1 - 3| = 2$$

Discrepancy on Set Families

- Informally, split some structure in a balanced way
- Fix some $\mathcal{F} \subseteq 2^{[n]}$ and let $\chi : [n] \rightarrow \{-1, 1\}$ be a two coloring
- Let R, B denote the set of units for each color, so $R = \chi^{-1}(-1), B = \chi^{-1}(1)$
- We say the discrepancy of $\mathcal{F}$ with respect to a coloring χ is

$$\text{disc}(\mathcal{F}, \chi) = \max_{S \in \mathcal{F}} ||S \cap R| - |S \cap B||$$



$$S_3 = [12] : \quad |S_3 \cap R| = 7, \quad |S_3 \cap B| = 5, \quad |7 - 5| = 2$$

Discrepancy on Set Families

- Informally, split some structure in a balanced way
- Fix some $\mathcal{F} \subseteq 2^{[n]}$ and let $\chi : [n] \rightarrow \{-1, 1\}$ be a two coloring
- Let R, B denote the set of units for each color, so $R = \chi^{-1}(-1), B = \chi^{-1}(1)$
- We say the discrepancy of $\mathcal{F}$ with respect to a coloring χ is

$$\text{disc}(\mathcal{F}, \chi) = \max_{S \in \mathcal{F}} ||S \cap R| - |S \cap B||$$



$$\text{disc}(\mathcal{F}, \chi) = \max(3, 2, 2) = 3$$

- Because χ maps to $\{-1, 1\}$ we can write the imbalance as

$$\text{disc}(\mathcal{F}, \chi) = \max_{S \in \mathcal{F}} \left| \sum_{u \in S} \chi(u) \right|$$

Discrepancy on Set Families

- Because χ maps to $\{-1, 1\}$ we can write the imbalance as

$$\text{disc}(\mathcal{F}, \chi) = \max_{S \in \mathcal{F}} \left| \sum_{u \in S} \chi(u) \right|$$

- The above measures how balanced $\mathcal{F}$ is under χ , we say the discrepancy of $\mathcal{F}$ itself is

$$\text{disc}(\mathcal{F}) = \min_{\chi} \text{disc}(\mathcal{F}, \chi)$$

- If $\mathcal{F} = 2^{[n]}$ then $R, B \in \mathcal{F}$ where one must have $\geq n/2$ elements so $\text{disc}(\mathcal{F}) \geq n/2$

- If $\mathcal{F} = 2^{[n]}$ then $R, B \in \mathcal{F}$ where one must have $\geq n/2$ elements so $\text{disc}(\mathcal{F}) \geq n/2$
- If $\mathcal{F}$ is a collection of disjoint sets we can choose χ to guarantee $\text{disc}(\mathcal{F}) \leq 1$

- If $\mathcal{F} = 2^{[n]}$ then $R, B \in \mathcal{F}$ where one must have $\geq n/2$ elements so $\text{disc}(\mathcal{F}) \geq n/2$
- If $\mathcal{F}$ is a collection of disjoint sets we can choose χ to guarantee $\text{disc}(\mathcal{F}) \leq 1$
- We can use the probabilistic method to show $\text{disc}(\mathcal{F}) \leq \sqrt{n \log(2|\mathcal{F}|)}$

Observe that if χ is chosen uar, for any $S \in \mathcal{F}$ we can use Hoeffding's to show

$$\Pr \left[\left| \sum_{u \in S} \chi(u) \right| > t \right] \leq 2e^{-t^2/2|S|} \leq 2e^{-t^2/2n}$$

then we set $t = \sqrt{2n \ln(2|\mathcal{F}|)}$ so that the above is $< 1/|\mathcal{F}|$. From there you can union bound over every set $S \in \mathcal{F}$ to say that under χ all sets have the desired discrepancy with positive probability, implying existence.

□

- For some $\mathcal{F}$, collect incidence vectors along rows to form a matrix in $B \in \mathbb{R}^{m \times n}$

Matrix Discrepancy

- For some $\mathcal{F}$, collect incidence vectors along rows to form a matrix in $B \in \mathbb{R}^{m \times n}$
- Rows indexed by sets $S \in \mathcal{F}$, and columns index elements in $[m]$

$$\text{disc}(\mathcal{F}) = \text{disc}(B) = \min_{z \in \{-1, 1\}^n} \|Bz\|_{\infty}$$

Matrix Discrepancy

- For some $\mathcal{F}$, collect incidence vectors along rows to form a matrix in $B \in \mathbb{R}^{m \times n}$
- Rows indexed by sets $S \in \mathcal{F}$, and columns index elements in $[m]$

$$\text{disc}(\mathcal{F}) = \text{disc}(B) = \min_{z \in \{-1, 1\}^n} \|Bz\|_{\infty}$$

- Matrix Discrepancy: split the columns of B into two groups with similar sums

Matrix Discrepancy

- For some $\mathcal{F}$, collect incidence vectors along rows to form a matrix in $B \in \mathbb{R}^{m \times n}$
- Rows indexed by sets $S \in \mathcal{F}$, and columns index elements in $[m]$

$$\text{disc}(\mathcal{F}) = \text{disc}(B) = \min_{z \in \{-1, 1\}^n} \|Bz\|_{\infty}$$

- Matrix Discrepancy: split the columns of B into two groups with similar sums
- Random colorings are the baseline, can we do better

- Spencer, Six Standard Deviations Suffice (1985): For binary B where $n = m$ we have

$$\text{disc}(B) \leq 6\sqrt{n}$$

- Spencer, Six Standard Deviations Suffice (1985): For binary B where $n = m$ we have

$$\text{disc}(B) \leq 6\sqrt{n}$$

- Proof was nonconstructive, not known if such a coloring could be found in poly time

Key Results in Discrepancy

- Spencer, Six Standard Deviations Suffice (1985): For binary B where $n = m$ we have

$$\text{disc}(B) \leq 6\sqrt{n}$$

- Proof was nonconstructive, not known if such a coloring could be found in poly time
- Bansal (2010) showed such an algorithm exists

Key Results in Discrepancy

- Spencer, Six Standard Deviations Suffice (1985): For binary B where $n = m$ we have

$$\text{disc}(B) \leq 6\sqrt{n}$$

- Proof was nonconstructive, not known if such a coloring could be found in poly time
- Bansal (2010) showed such an algorithm exists
- Algorithmic discrepancy theory
 - Applications: attention approximation, experimental design, monte carlo integration

Recent Results

- Discrepancy theory has had a lot of recent breakthroughs discovered by AI
- Guo, Fang, Lu (Sep. 2026) showed for B with column norm ≤ 1 (Komlós problem)
- $$\text{disc}(B) = O(1)$$
 - Li 2026 shows algorithm such a coloring in $O(mn + n^{\omega+2} \log^3 n)$ time
 - Bansal, Jiang 26: $\tilde{O}((\log n)^{1/4})$
 - Banaszczyk 98: $O(\sqrt{\log n})$
- Aden-Ali 2026: Optimal algorithm for online vector balancing

The Gram-Schmidt Walk



The Gram-Schmidt Walk

- Due to Bansal, Dadush, Lovett, Garg in 2018

The Gram-Schmidt Walk

- Due to Bansal, Dadush, Lovett, Garg in 2018
- Shows that we can sample $z \in \{-1, 1\}^n$ such that Bz is $O(1)$ -Subgaussian

The Gram-Schmidt Walk

- Due to Bansal, Dadush, Lovett, Garg in 2018
- Shows that we can sample $z \in \{-1, 1\}^n$ such that Bz is $O(1)$ -Subgaussian
 - Implies $\text{disc}(B) = O(\sqrt{\log n})$

The Gram-Schmidt Walk

- Due to Bansal, Dadush, Lovett, Garg in 2018
- Shows that we can sample $z \in \{-1, 1\}^n$ such that Bz is $O(1)$ -Subgaussian
 - Implies $\text{disc}(B) = O(\sqrt{\log n})$
 - A random variable X σ^2 -subgaussian if $\forall t \in \mathbb{R}$

$$\mathbb{E}[e^{tX}] \leq e^{\frac{1}{2}t^2\sigma^2}$$

- and for random vectors $y \in \mathbb{R}^m$ we require $\forall v \in \mathbb{R}^m$

$$\mathbb{E}[e^{\langle y, v \rangle}] \leq e^{\frac{1}{2}\|v\|^2\sigma^2}$$

The Gram-Schmidt Walk

- Due to Bansal, Dadush, Lovett, Garg in 2018
- Shows that we can sample $z \in \{-1, 1\}^n$ such that Bz is $O(1)$ -Subgaussian
 - Implies $\text{disc}(B) = O(\sqrt{\log n})$
 - A random variable X σ^2 -subgaussian if $\forall t \in \mathbb{R}$

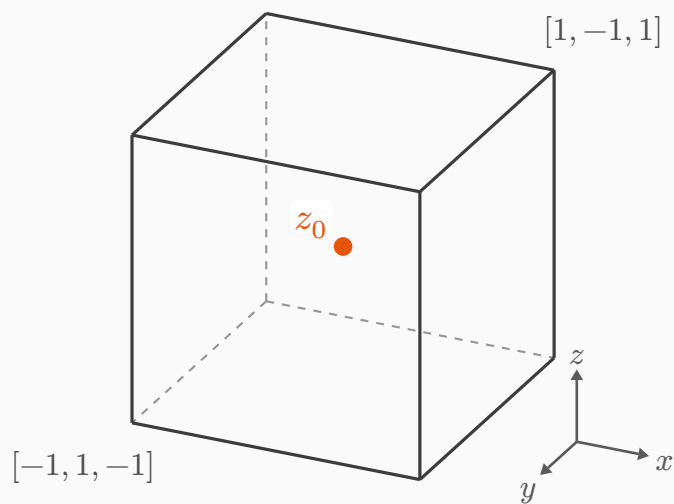
$$\mathbb{E}[e^{tX}] \leq e^{\frac{1}{2}t^2\sigma^2}$$

- and for random vectors $y \in \mathbb{R}^m$ we require $\forall v \in \mathbb{R}^m$

$$\mathbb{E}[e^{\langle y, v \rangle}] \leq e^{\frac{1}{2}\|v\|^2\sigma^2}$$

- Walk around in $[-1, 1]^n$ randomly taking steps with low discrepancy wrt B

Visualization



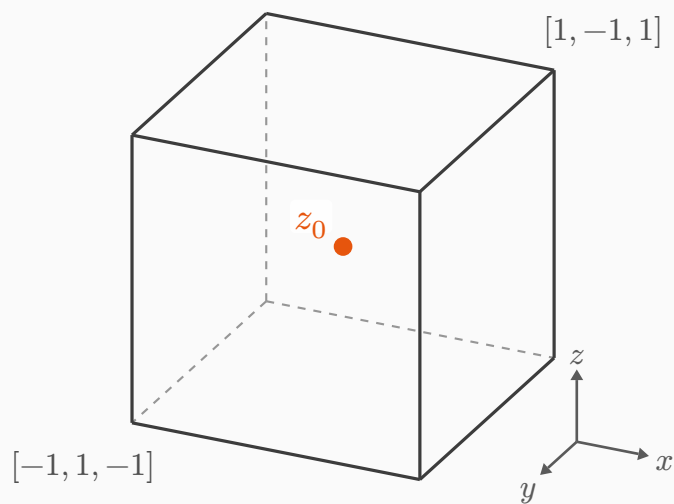
Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3
$B =$	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0

$z_0 = [0, 0, 0]$
 $A_1 = \{1, 2, 3\}$

Visualization



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

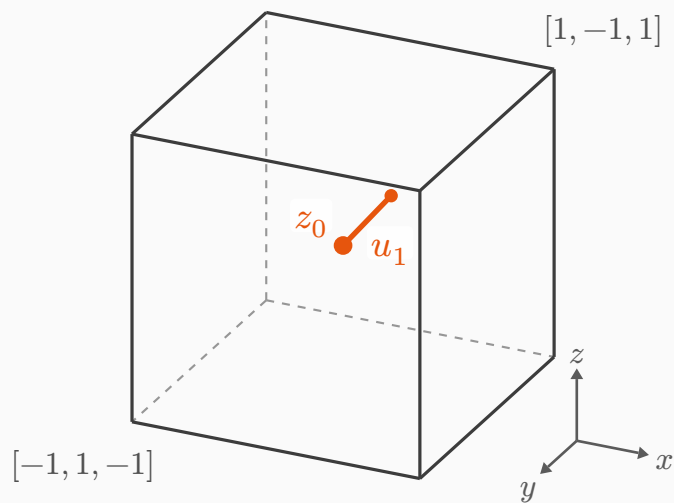
- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3
$B =$	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0

$$z_0 = [0, 0, 0]$$

$$A_1 = \{1, 2, 3\}, \quad p_1 = 3$$

Visualization



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3
	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
$B =$	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0
$u_1 =$	0.82	0.2	1

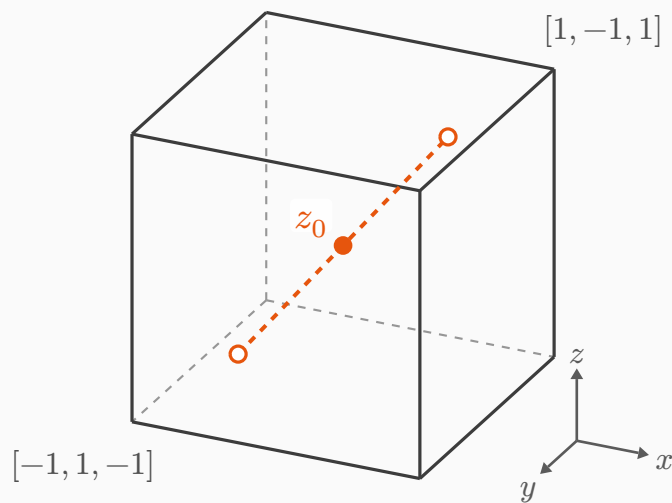
$$Bu_1 = v_3 + 0.82v_1 + 0.2v_2 =$$

$$[0.69, 0.05, 0.14, -0.37, 0.23, -0.02]$$

$$z_0 = [0, 0, 0]$$

$$A_1 = \{1, 2, 3\}, \quad p_1 = 3$$

Visualization



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3
	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
$B =$	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0
$u_1 =$	0.82	0.2	1

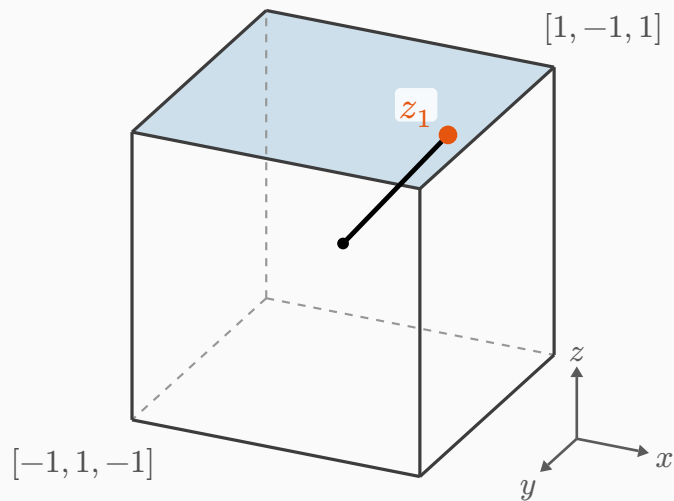
$$Bu_1 = v_3 + 0.82v_1 + 0.2v_2 =$$

$$[0.69, 0.05, 0.14, -0.37, 0.23, -0.02]$$

$$\delta_1^+ = 1, \quad \delta_1^- = 1$$

$$\Pr[+] = 0.5, \quad \Pr[-] = 0.5$$

Visualization



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

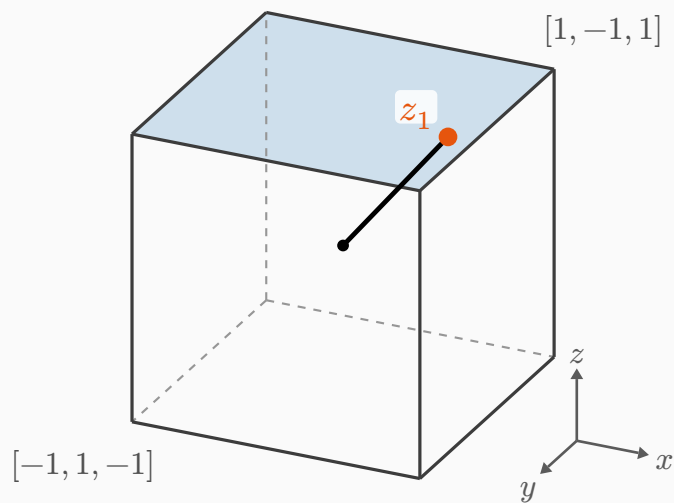
- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3
$B =$	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0

$$\delta_1 = \delta_1^+ : z_1 = [0.82, 0.2, 1]$$

$$A_2 = \{1, 2\}$$

Visualization



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

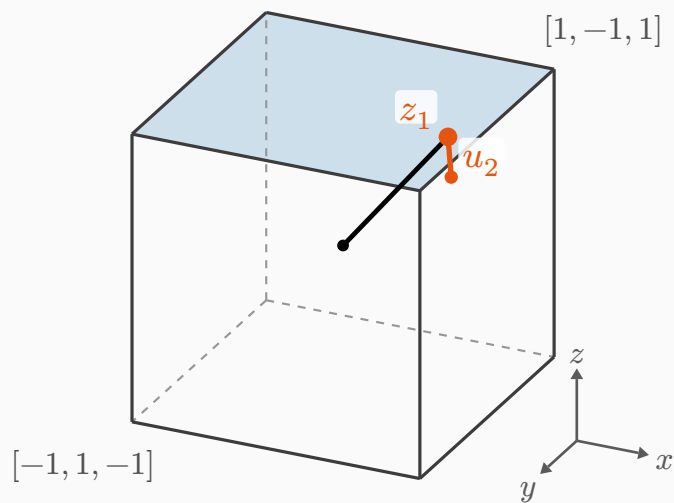
- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3
$B =$	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0

$$z_1 = [0.82, 0.2, 1]$$

$$A_2 = \{1, 2\}, \quad p_2 = 2$$

Visualization



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3
	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
$B =$	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0
$u_2 =$	0.51	1	0

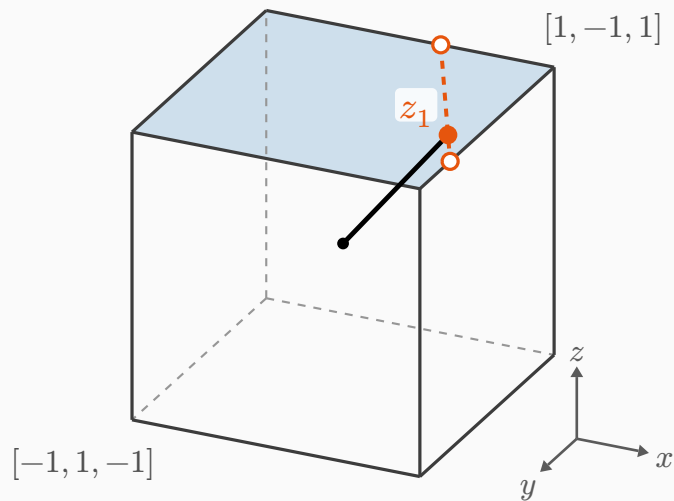
$$Bu_2 = v_2 + 0.51v_1 =$$

$$[-0.1, -0.5, 0.35, -0.05, 0.07, -0.36]$$

$$z_1 = [0.82, 0.2, 1]$$

$$A_2 = \{1, 2\}, p_2 = 2$$

Visualization



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3
$B =$	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0
$u_2 =$	0.51	1	0

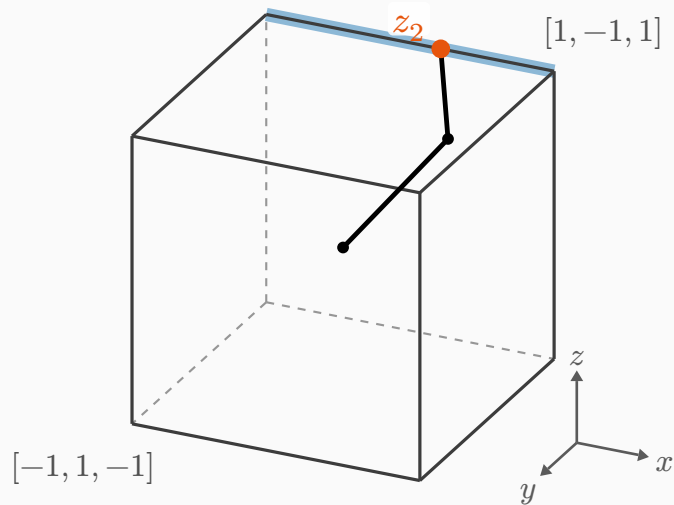
$$Bu_2 = v_2 + 0.51v_1 =$$

$$[-0.1, -0.5, 0.35, -0.05, 0.07, -0.36]$$

$$\delta_2^+ = 0.35, \quad \delta_2^- = 1.2$$

$$\Pr[+] = 0.77, \quad \Pr[-] = 0.23$$

Visualization



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

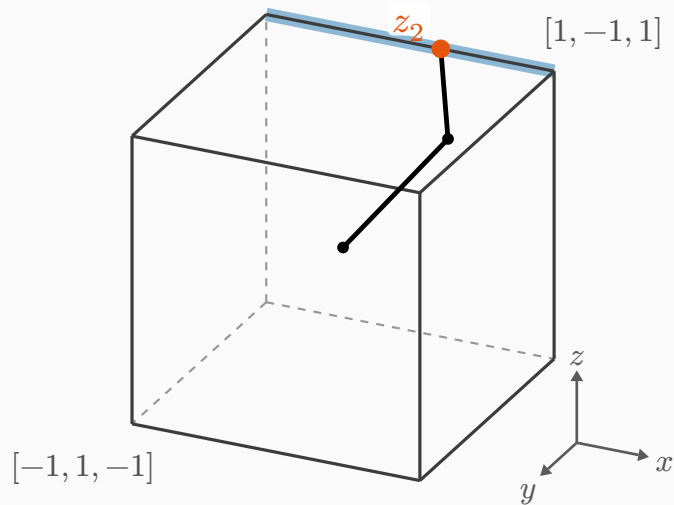
- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3
$B =$	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0

$$\delta_2 = -\delta_2^-: z_2 = [0.21, -1, 1]$$

$$A_3 = \{1\}$$

Visualization



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

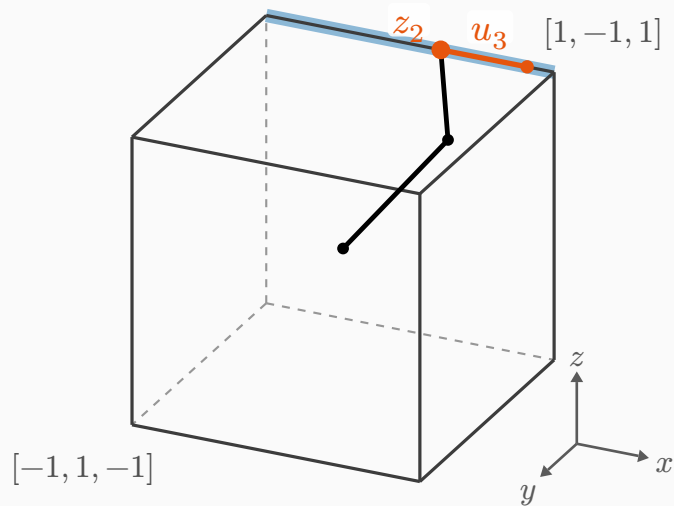
- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

v_1	v_2	v_3
-0.4	0.1	1
0.2	-0.6	0
0.1	0.3	0
-0.5	0.2	0
0.3	-0.08	0
0.07	-0.4	0

$$z_2 = [0.21, -1, 1]$$

$$A_3 = \{1\}, \quad p_3 = 1$$

Visualization



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

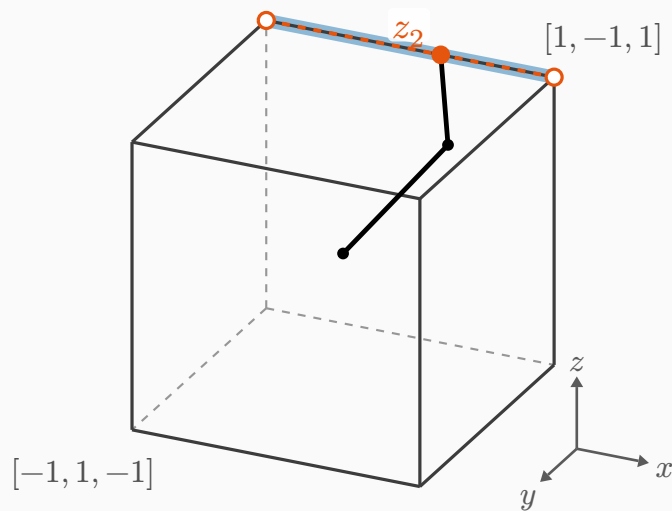
	v_1	v_2	v_3
$B =$	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0
$u_3 =$	1	0	0

$$Bu_3 = v_1 = [-0.4, 0.2, 0.1, -0.5, 0.3, 0.07]$$

$$z_2 = [0.21, -1, 1]$$

$$A_3 = \{1\}, p_3 = 1$$

Visualization



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

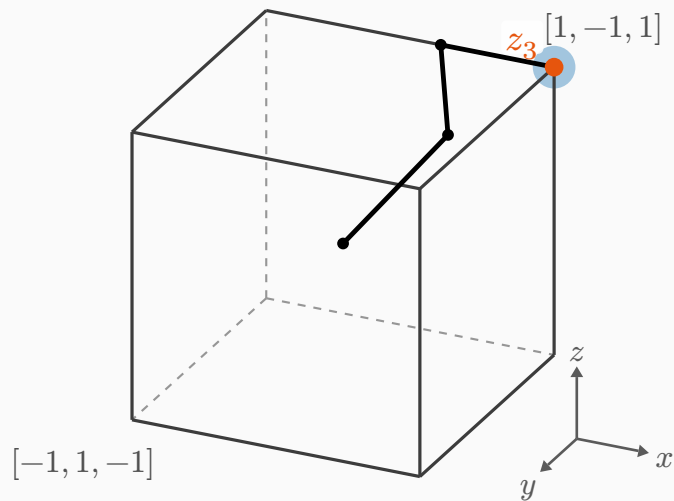
	v_1	v_2	v_3
$B =$	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0
$u_3 =$	1	0	0

$$Bu_3 = v_1 = [-0.4, 0.2, 0.1, -0.5, 0.3, 0.07]$$

$$\delta_3^+ = 0.79, \quad \delta_3^- = 1.21$$

$$\Pr[+] = 0.61, \quad \Pr[-] = 0.39$$

Visualization



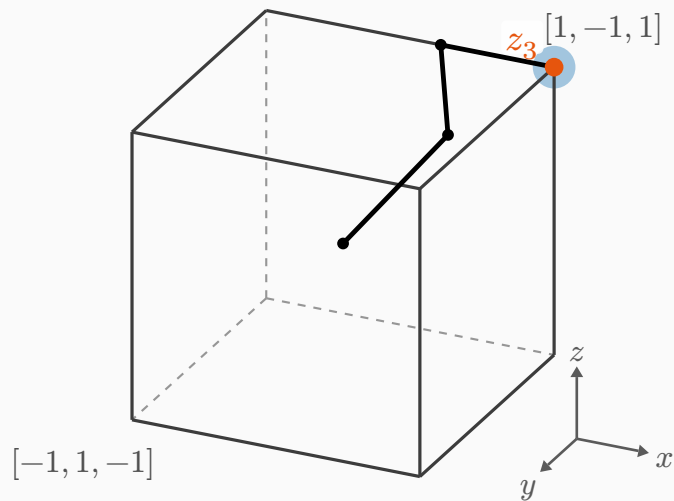
Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3
	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
$B =$	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0

$$\delta_3 = \delta_3^+ : z_3 = [1, -1, 1]$$
$$A_4 = \emptyset$$

Visualization



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3
	-0.4	0.1	1
	0.2	-0.6	0
	0.1	0.3	0
$B =$	-0.5	0.2	0
	0.3	-0.08	0
	0.07	-0.4	0

Done: coloring
 $\chi = [1, -1, 1]$

- Take advantage of linear dependencies between columns, but stay sufficiently random

The Gram-Schmidt Walk

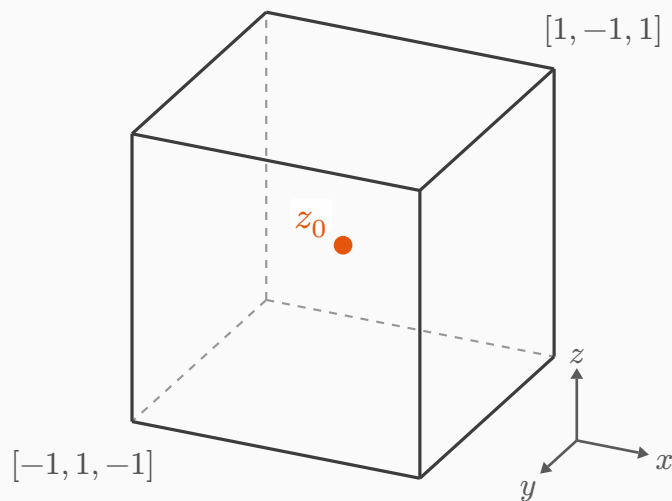
- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



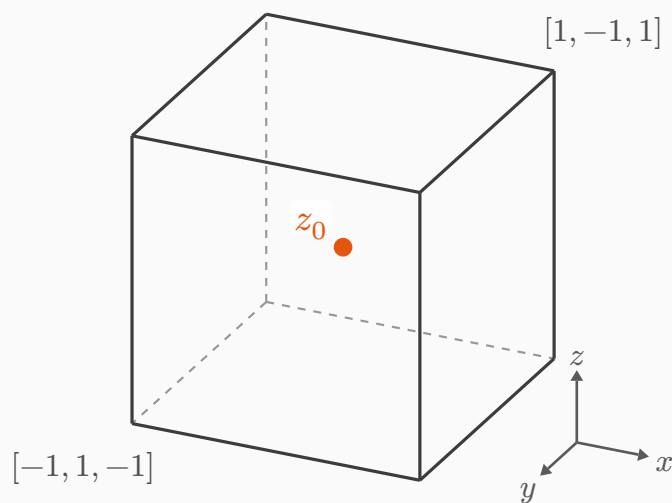
Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3	
	1	0	0	
$B =$	0	1	0	$z_0 = [0, 0, 0]$
	0	0	1	$A_1 = \{1, 2, 3\}$

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



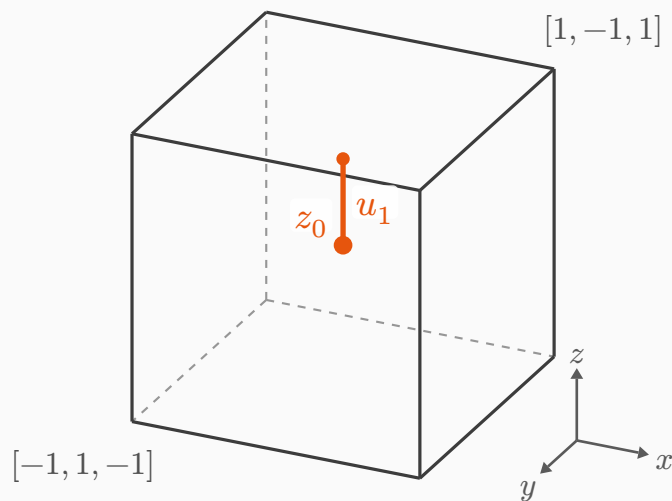
Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3	
$B =$	1	0	0	$z_0 = [0, 0, 0]$
	0	1	0	$A_1 = \{1, 2, 3\}, p_1 = 3$
	0	0	1	

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

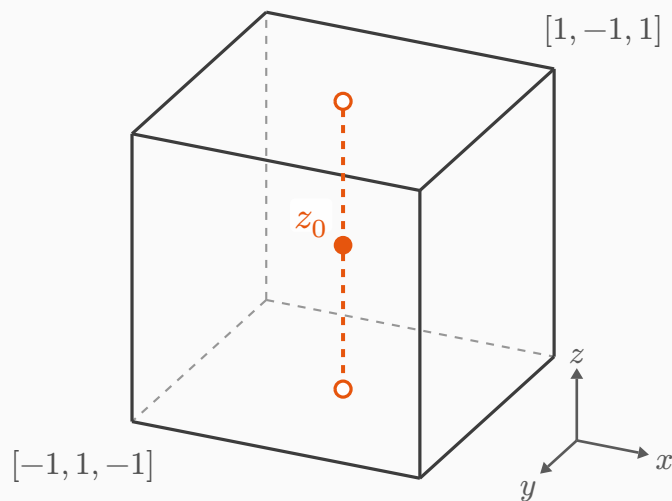
- $p_t \leftarrow \max A_t$ (pivot)
- choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

	v_1	v_2	v_3	
	1	0	0	
$B =$	0	1	0	
	0	0	1	
$u_1 =$	0	0	1	

$Bu_1 = v_3 + 0v_1 + 0v_2 = [0, 0, 1]$
 $z_0 = [0, 0, 0]$
 $A_1 = \{1, 2, 3\}, p_1 = 3$

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- $p_t \leftarrow \max A_t$ (pivot)
- choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

$$B = \begin{array}{c|cc} & v_1 & v_2 & v_3 \\ \hline u_1 = & 1 & 0 & 0 \\ & 0 & 1 & 0 \\ & 0 & 0 & 1 \end{array}$$

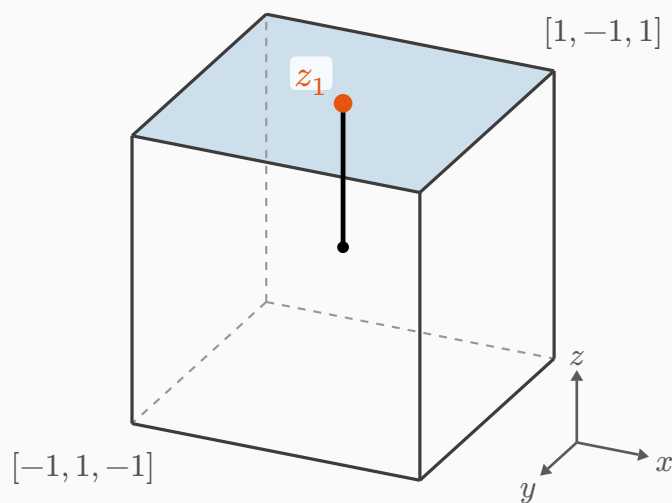
$$Bu_1 = v_3 + 0v_1 + 0v_2 = [0, 0, 1]$$

$$\delta_1^+ = 1, \delta_1^- = 1$$

$$\Pr[+] = 0.5, \Pr[-] = 0.5$$

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- $p_t \leftarrow \max A_t$ (pivot)
- choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

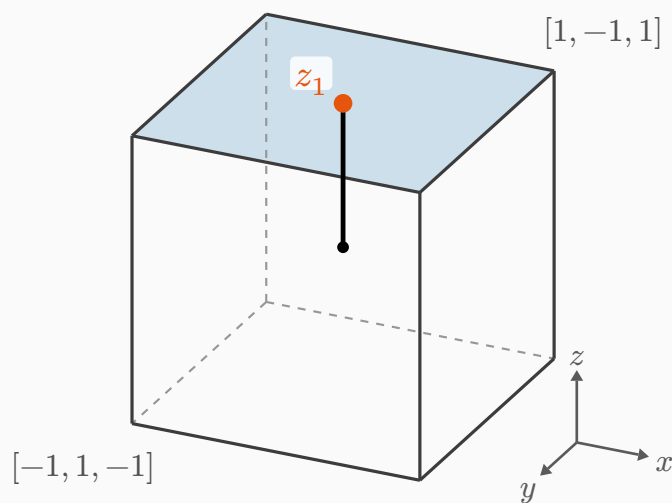
	v_1	v_2	v_3
$B =$	1	0	0
	0	1	0
	0	0	1

$$\delta_1 = \delta_1^+ : z_1 = [0, 0, 1]$$

$$A_2 = \{1, 2\}$$

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

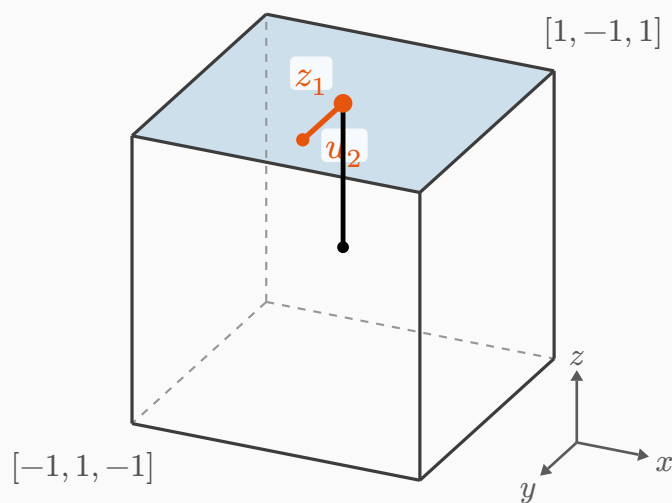
	v_1	v_2	v_3
$B =$	1	0	0
	0	1	0
	0	0	1

$$z_1 = [0, 0, 1]$$

$$A_2 = \{1, 2\}, \quad p_2 = 2$$

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- $p_t \leftarrow \max A_t$ (pivot)
- choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

$$B = \begin{array}{c|ccc} & v_1 & v_2 & v_3 \\ \hline & 1 & 0 & 0 \\ & 0 & 1 & 0 \\ & 0 & 0 & 1 \end{array}$$

$$u_2 = \begin{array}{c|ccc} & v_1 & v_2 & v_3 \\ \hline & 0 & 1 & 0 \end{array}$$

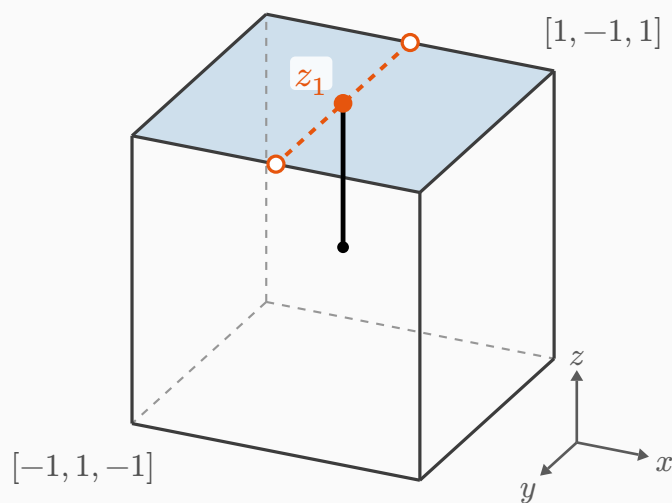
$$Bu_2 = v_2 + 0v_1 = [0, 1, 0]$$

$$z_1 = [0, 0, 1]$$

$$A_2 = \{1, 2\}, \quad p_2 = 2$$

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- $p_t \leftarrow \max A_t$ (pivot)
- choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

$$B = \begin{array}{c|ccc} & v_1 & v_2 & v_3 \\ \hline & 1 & 0 & 0 \\ & 0 & 1 & 0 \\ & 0 & 0 & 1 \end{array}$$

$$u_2 = \begin{array}{c|ccc} & v_1 & v_2 & v_3 \\ \hline & 0 & 1 & 0 \end{array}$$

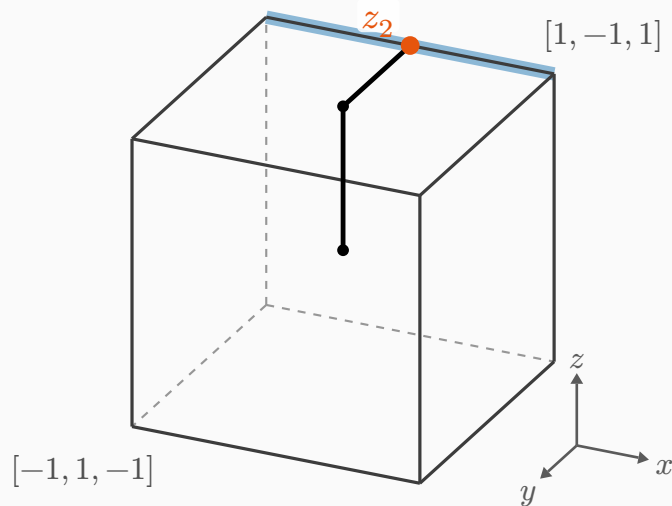
$$Bu_2 = v_2 + 0v_1 = [0, 1, 0]$$

$$\delta_2^+ = 1, \quad \delta_2^- = 1$$

$$\Pr[+] = 0.5, \quad \Pr[-] = 0.5$$

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- $p_t \leftarrow \max A_t$ (pivot)
- choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

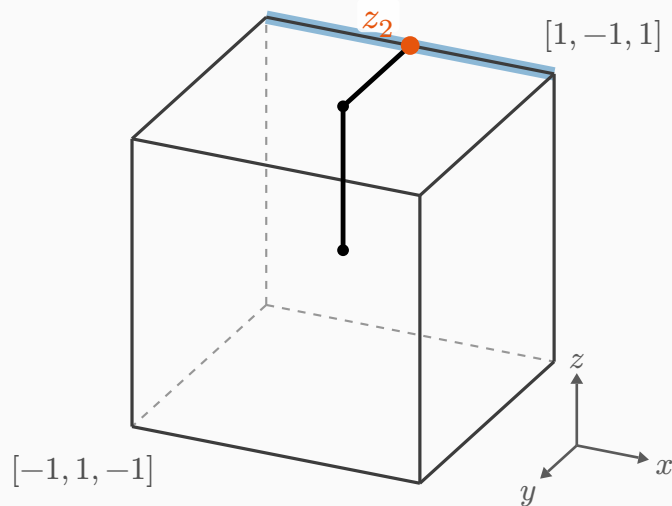
	v_1	v_2	v_3
$B =$	1	0	0
	0	1	0
	0	0	1

$$\delta_2 = -\delta_2^-: z_2 = [0, -1, 1]$$

$$A_3 = \{1\}$$

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- $p_t \leftarrow \max A_t$ (pivot)
- choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

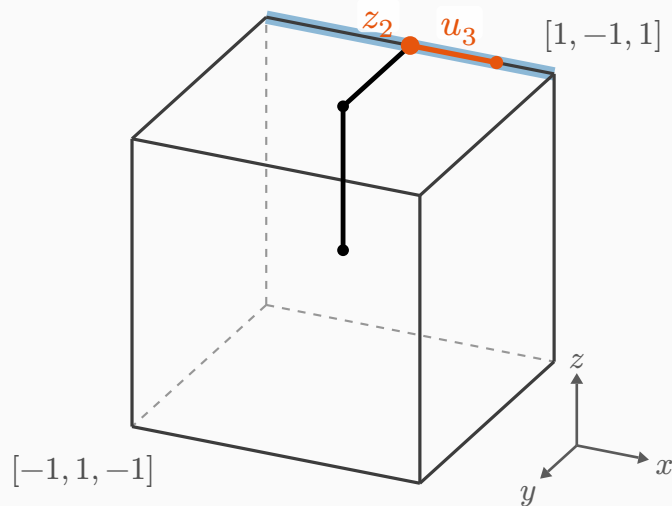
	v_1	v_2	v_3
$B =$	1	0	0
	0	1	0
	0	0	1

$$z_2 = [0, -1, 1]$$

$$A_3 = \{1\}, p_3 = 1$$

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- $p_t \leftarrow \max A_t$ (pivot)
- choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

$$B = \begin{array}{c|ccc} & v_1 & v_2 & v_3 \\ \hline u_3 = & \begin{array}{c} 1 \\ 0 \\ 0 \end{array} & \begin{array}{c} 0 \\ 1 \\ 0 \end{array} & \begin{array}{c} 0 \\ 0 \\ 1 \end{array} \end{array}$$

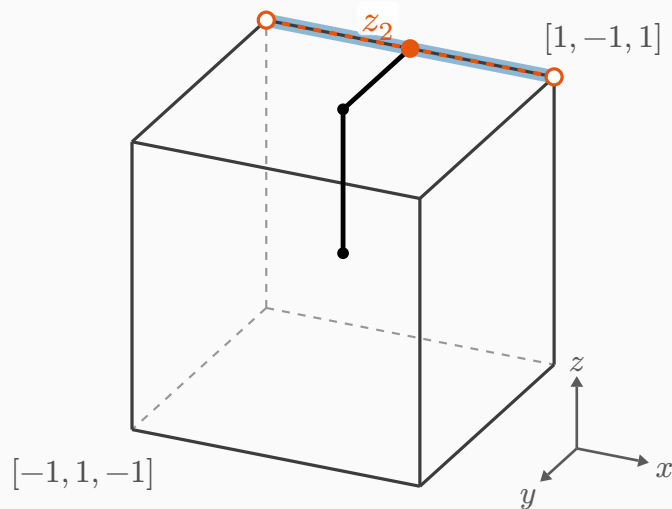
$$Bu_3 = v_1 = [1, 0, 0]$$

$$z_2 = [0, -1, 1]$$

$$A_3 = \{1\}, p_3 = 1$$

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- $p_t \leftarrow \max A_t$ (pivot)
- choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

$$B = \begin{array}{c|ccc} & v_1 & v_2 & v_3 \\ \hline & 1 & 0 & 0 \\ & 0 & 1 & 0 \\ & 0 & 0 & 1 \end{array}$$

$$u_3 = \begin{array}{c|ccc} & v_1 & v_2 & v_3 \\ \hline & 1 & 0 & 0 \end{array}$$

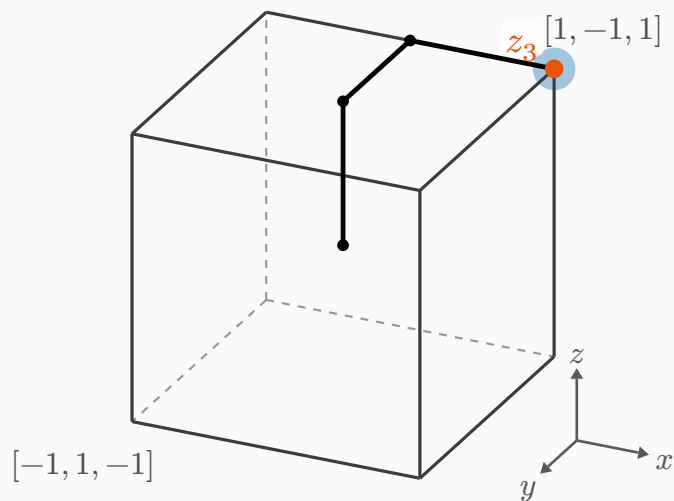
$$Bu_3 = v_1 = [1, 0, 0]$$

$$\delta_3^+ = 1, \delta_3^- = 1$$

$$\Pr[+] = 0.5, \Pr[-] = 0.5$$

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

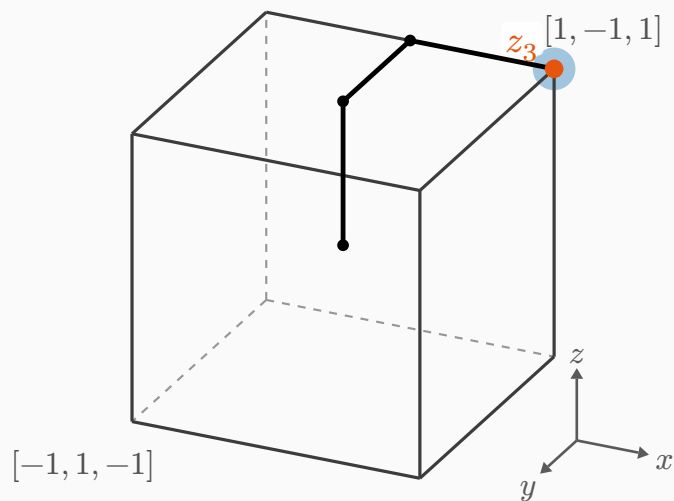
- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

$$B = \begin{array}{ccc} & v_1 & v_2 & v_3 \\ \begin{array}{c} 1 \\ 0 \\ 0 \end{array} & \begin{array}{c} 1 \\ 0 \\ 0 \end{array} & \begin{array}{c} 0 \\ 1 \\ 0 \end{array} & \begin{array}{c} 0 \\ 0 \\ 1 \end{array} \end{array}$$

$$\delta_3 = \delta_3^+ : z_3 = [1, -1, 1] \\ A_4 = \emptyset$$

The Gram-Schmidt Walk

- Take advantage of linear dependencies between columns, but stay sufficiently random
- When $B = I_n$ we get z is i.i.d uniform in each entry
- When B is a single column repeated we get perfect cancellations, $Bz = 0$
- Linear dependencies correspond to negative correlations in z



Start at $z_0 = 0$ with $A_1 = [n]$. While $A_t \neq \emptyset$:

- 1 $p_t \leftarrow \max A_t$ (pivot)
- 2 choose u_t to minimize $\|Bu_t\|$ such that $u_t(p_t) = 1$ and $u_t(i) = 0$ for $i \notin A_t$
- 3 let $\delta_t^+, \delta_t^- > 0$ be the step sizes at which $z_{t-1} + \delta_t^+ u_t$ and $z_{t-1} - \delta_t^- u_t$ hit a face
- 4 take a step $z_t = z_{t-1} + \delta_t u_t$, where $\delta_t \in \{\delta_t^+, -\delta_t^-\}$ is chosen with mean zero
- 5 update the alive set: $A_{t+1} = \{i : |z_t(i)| < 1\}$

$B =$	v_1	v_2	v_3
	1	0	0
	0	1	0
	0	0	1

Done: coloring
 $\chi = [1, -1, 1]$

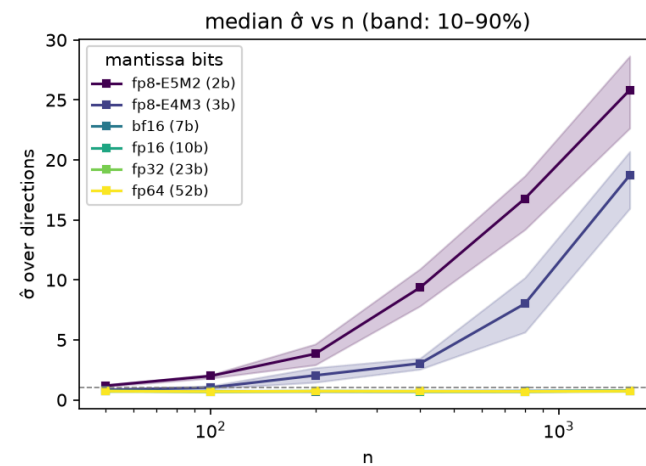
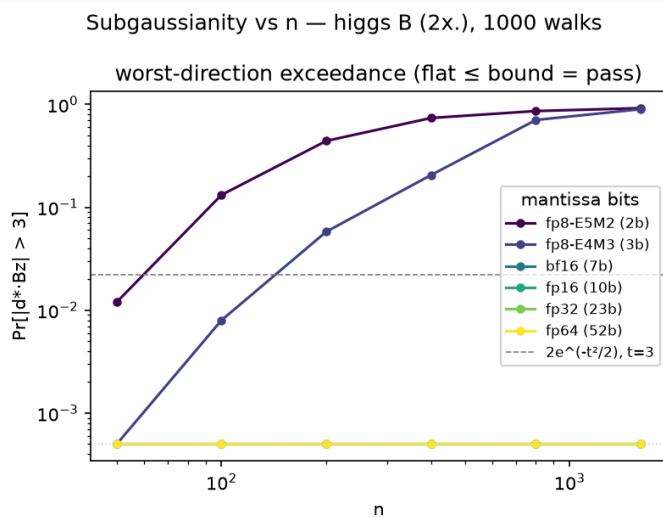
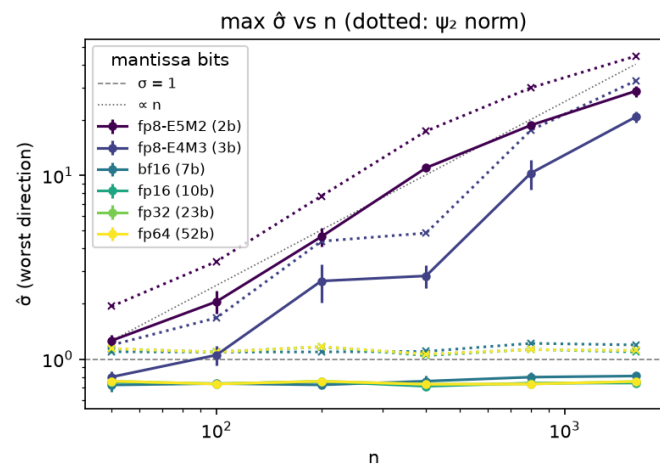
- Harshaw, Savje, Spielman, Zhang 19: Bz is 1-Subgaussian
 - Motivated by experimental design applications

Our Contributions



- Determining u_t at each iteration requires solving a least squares system
- In practice: speed this up by maintaining some factorization to be reused at every t
 - Errors accumulate over iterations
 - Approximate u_t can change which units get decided, so trajectory differs
- How does approximate u_t hurt our performance

Experiments



Theoretical Result

$$\mathbb{E}[\exp(\langle Bz, v \rangle)] \leq \exp \left(\frac{1}{2} \|v\|^2 + \left(\underbrace{n^{3/2}(2\varepsilon_{\text{pr}} + \varepsilon_{\text{iter}})}_{\text{drift/sampling}} + \left(\underbrace{2n\varepsilon_{\text{LS}}}_{\text{least squares}} + \underbrace{n(n+1)\varepsilon_{\text{iter}}}_{\text{arithmetic/clamping}} \right) \right) \|v\| \right)$$

- Roughly speaking the 1-subgaussianity is more like $1 + O(n^2\varepsilon)$

Theoretical Result

$$\mathbb{E}[\exp(\langle Bz, v \rangle)] \leq \exp \left(\frac{1}{2} \|v\|^2 + \left(\underbrace{n^{3/2}(2\varepsilon_{\text{pr}} + \varepsilon_{\text{iter}})}_{\text{drift/sampling}} + \left(\underbrace{2n\varepsilon_{\text{LS}}}_{\text{least squares}} + \underbrace{n(n+1)\varepsilon_{\text{iter}}}_{\text{arithmetic/clamping}} \right) \right) \|v\| \right)$$

- Roughly speaking the 1-subgaussianity is more like $1 + O(n^2\varepsilon)$
- As $\varepsilon \rightarrow 0$ this recovers the original result

Theoretical Result

$$\mathbb{E}[\exp(\langle Bz, v \rangle)] \leq \exp \left(\frac{1}{2} \|v\|^2 + \left(\underbrace{n^{3/2}(2\varepsilon_{\text{pr}} + \varepsilon_{\text{iter}})}_{\text{drift/sampling}} + \left(\underbrace{2n\varepsilon_{\text{LS}}}_{\text{least squares}} + \underbrace{n(n+1)\varepsilon_{\text{iter}}}_{\text{arithmetic/clamping}} \right) \right) \|v\| \right)$$

- Roughly speaking the 1-subgaussianity is more like $1 + O(n^2\varepsilon)$
- As $\varepsilon \rightarrow 0$ this recovers the original result
- $\varepsilon_{\text{pr}}, \varepsilon_{\text{iter}}$ scale like $\varepsilon_{\text{mach}}$, ε_{LS} scales like $\kappa\varepsilon_{\text{mach}}$ if solver is backwards stable

Conclusion



- Our proofs rely heavily on boundedness
 - Distributional assumptions on rounding?

- Our proofs rely heavily on boundedness
 - Distributional assumptions on rounding?
- Precise result for particular solvers?
 - Would require characterizing stability of least squares solvers under rank one updates

- Our proofs rely heavily on boundedness
 - Distributional assumptions on rounding?
- Precise result for particular solvers?
 - Would require characterizing stability of least squares solvers under rank one updates
- Can we explain this sharp transition behavior where 30 or so bits seems to be indistinguishable from 64